The initial step of replicating the computer lab within Unity and Blender was to thoroughly measure important aspects of the lab. The objects include desks, walls, floors, and other large static objects and equipment. From these measurements, Unity and Blender was used to digitally replicate the environment.

Blender was used to design finer details of the lab's equipment. Specifically objects that are not easily created with Unity's 3d Object selection. In Blender, irregular objects can be customized and designed to scale and easily imported into the Unity scene.

In Unity, spatial layout of the lab was important. Meticulously aligning, virtual walls, desks, and equipment to mirror the physical space. All objects are then combined together in the Unity scene in a one to one scale. All objects must be accurately represented to allow for accurate testing. Testing the Unity space in virtual reality is important for making adjustments to perfect the layout of the Environment.

This virtual environment is necessary for testing of the quizzing and other subsequent features. Virtual reality is also easier to test versus an Augmented Reality system. But this complete system should be convertible into an Augmented Reality application. When the user enters the system, the objects in virtual environment should be accurately depict their environment.

Objects recognized with the Yolov8 system will be assigned a prefab. A prefab is short for a 'prefabricated object' and it is a reusable asset complete with its own associated properties, scripts, and behaviors and can be instantiated multiple times throughout the application. The prefab contains multiple elements that will aid the user in learning foreign words. The prefab includes a in-situ text label that can be converted from the player's native language to their target foreign language.

In order to generate words into their corresponding foreign language a dictionary can be written into a script. Words from a CSV (comma-separated values) file can be used in this dictionary. The CSV file will be set up so that the native words are placed in the first column (key) and the foreign words are placed in the second column(value).
Figure 2: A visualization of the Unity Scene Setup

3.3 Learning System

The learning component of the system draws from Anki learning system, which operates on the foundational principles of spaced repetition. Anki's methodology revolves around strategically timed review sessions aimed at optimizing the long-term retention of information.

To adapt this system into a different learning environment, we propose a method where users are presented with a word displayed above an object. By clicking on the displayed word, users can toggle it to reveal the corresponding foreign word, fostering an interactive and an immersive learning experience. The displaying of words above objects has taken inspiration from work done in the ARbis Pictus paper [2]. In the ARbis Pictus system, corresponding foreign words are placed above the objects with an audio recording of that object.

Figure 3: A visualization of the text label above the simulated object

Following this interaction, users are prompted with three distinct buttons, each representing varying degrees of understanding: ”Don't Know”, ”Recognized”, and ”Know”. These buttons emulate the feedback mechanism in the Anki system, allowing users to assess their comprehension level of the presented material. Depending on the button clicked, the system dynamically adjusts the visibility of the object's label, temporarily hiding it for a specified duration before reintroducing it into the learning environment. Notably, the duration of concealment increases incrementally as users progress from indicating unfamiliarity, ”Don't Know”, to familiarity ,”Know”.

Figure 4: A visualization of the language translation and the knowledge rating buttons. These are presented when the text label button is pressed (see Figure 2)

3.4 Quizzing System

Within the learning system, an additional feature enriching the user experience is the incorporation of a multiple-choice quiz. This interactive assessment tool serves as a means for users to gauge their understanding against a broader spectrum of vocabulary words. The decision to implement a multiple-choice format stems from its simplicity and effectiveness. Drawing upon the data stored in the CSV file, the quiz dynamically selects words that share the same initial letter as the target word. This strategic selection not only enriches the learning experience but also introduces an element of challenge. The versatility of this quiz can extend beyond the vocabulary assessment. It can be adapted for practical applications such as fill-in-the-blank exercises. By presenting users with a selection of answer choices, each representing potential completions for a given sentence or phrase, the quiz becomes a versatile tool for reinforcing contextual understanding and language proficiency. The multiple choice quiz can be combined

with the Anki learning system as a way to either increase or decrease
the time between hiding and showing words.
Figure 5: A visualization of the quizzing system. Result are located above the quiz.